CFPNet: Coarse-to-Fine Progressive Network for Cloud Detection in Remote Sensing Images

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***Abstract*—Accurate cloud detection in remote sensing images constitutes a critical preprocessing requirement for ensuring the validity of subsequent analytical applications. Existing cloud detection methods effectively identify primary cloud regions but struggle with precise boundary delineation and distinguishing spectrally similar surfaces, particularly between clouds and terrain with comparable reflectance properties. To address these challenges, we propose the coarse-to-fine progressive network (CFPNet), a progressive detection framework integrating across channel and spatial dimensions. The framework incorporates two innovative components: a channel spatial attention residual fusion (CSARF) module and a multimask adaptive attention (MMAA) module. The CSARF module achieves the network's initial focus on clouds, while the MMAA module enables fine feature extraction of clouds. Specifically, The CSARF module uses channel attention module (CAM) and spatial attention module (SAM) to suppress background noise while enhancing discriminative feature representation. MMAA uses multimask self-attention (MMSA) to compute channel self-attention and capture long-range dependencies, while using a multimask strategy to filter important channels. Deformable contextual feedforward network (DCFN) then adaptively extracts cloud boundary features through deformable convolution, minimizing noncloud pixel interference. Hence, the network enables coarse-to-fine feature extraction of clouds across both channel and spatial dimensions. Experimental results on the GF1-WFV, AIR-CD, and Sentinel-2 datasets demonstrate that our method achieves superior performance in boundary detail accuracy and interclass feature classification compared with other methods.**

***Index Terms*—Cloud detection, deep learning, multigranular attention, remote sensing image.**

I. Introduction

With the development of remote sensing technology, remote sensing images have been widely applied in numerous scenes [1], [2], [3], [4], such as water resource detection, environmental pollution monitoring, urban planning, and postdisaster assessment. Zhang et al. [5] indicate that approximately 66% of the Earth's surface is typically covered by clouds. This extensive cloud coverage can obscure optical remote sensing images, leading to significant loss of feature information and reduced data availability [6]. Cloud detection techniques, which can identify cloud regions in optical remote sensing images, provide a basis for evaluating the data quality of satellite-transmitted optical remote sensing images. Thus, researching accurate cloud detection methods holds great significance.

In recent years, researchers have proposed numerous cloud detection methods, which can be broadly categorized into three classes: threshold-based, spatial-feature-based, and deep-learning-based methods. Threshold-based methods primarily rely on analyzing the spectral characteristics of each remote sensing image band and setting thresholds to distinguish between cloud pixels and noncloud pixels. For example, Zhu et al. [7] introduced Fmask for Landsat scenes through spectral thresholds. Li et al. [8] introduced HYTA that combines fixed and adaptive thresholds. Li et al. [9] proposed MFC, refining a guided-filtered spectral mask with geometric textural features. Fisher [10] proposed SPOTCASM for cloud and cloud shadow detection in SPOT5 HRG images, combining 2-D histogram thresholding with morphological watershed segmentation and incorporating spectral, geometric, and area features to improve accuracy. Although effective in specific scenes, these methods remain sensitive to atmospheric conditions, reflectance, brightness, and temperature.

Spatial-feature-based methods involve manually designing and extracting specific features based on the physical properties of clouds (such as texture, color, and shape) and their spectral characteristics (including spectral reflectance and brightness temperature), and then using these features to classify cloud and noncloud regions. For instance, Kang et al. [11] proposed an unsupervised coarse-to-fine cloud detection method for remote sensing images that integrates color, texture, and statistical features with SVM classification and guided filter refinement. Ghasemian and Akhoondzadeh [12] introduced two methods that combine spectral features (e.g., NDVI, NDSI) and texture features using random forests and apply multiclassifier voting to distinguish clouds, snow, and background. Lin et al. [13] extracted radiometrically invariant pixels through weighted PCA, and then combined cloud-free image fusion with spectral disparity thresholding

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for cloud detection. Despite their strong classification performance, these methods rely on manual feature design. Given the diverse environments captured in remote sensing images, such as thin clouds whose features are easily influenced by the surfaces they cover, and the difficulty in distinguishing clouds from features with similar spectral characteristics (e.g., snow and ice), it remains challenging to develop a universal feature expression for cloud detection across different scenes [14].

The rapid advancement of deep learning has led to the emergence of many convolutional neural networks (CNNs), which have achieved outstanding results in semantic segmentation of natural images [15], [16], [17], [18], [19], [20], [21]. Given their strong feature modeling capability, CNNs have also been widely applied to remote sensing image semantic segmentation [22], [23], [24], [25], [26], [27]. Cloud detection, a crucial subtask in this field, aims to identify cloud-covered regions from multispectral or varying resolution images. However, due to the diverse shapes of clouds and spectral similarity with other terrains, traditional semantic segmentation methods often face challenges in direct application. Thus, designing targeted optimization strategies based on the architectural characteristics of CNNs has become pivotal for enhancing cloud detection accuracy. Dev et al. [28] introduced CloudSegNet, which uses multilevel feature extraction and segmentation mask reconstruction to achieve high performance with few parameters. Yan et al. [29] proposed MFFSNet, a multilevel feature fusion segmentation network for end-to-end cloud and cloud shadow detection. Its pyramid pooling module captures contextual associations between clouds and shadows, improving multiscale target segmentation. Yang et al. [30] proposed CDnet for cloud detection in low-resolution and incomplete spectral scenes. CDnet combines a feature pyramid module for multiscale information extraction and a boundary refinement module for optimizing cloud boundaries. Zhao et al. [31] introduced DMNet, incorporating a dense feature enhancement module to strengthen encoder features and a multiscale contextual fusion spatial attention module (SAM) to capture contextual details across scales, enhancing spatial features. Wu et al. [32] presented BoundaryNet, a dual-network architecture with scalable and differentiable boundary networks. The scalable network generates multiscale cloud masks, while the differentiable network refines boundaries using residual structures, with fused outputs producing precise cloud masks. Zhao et al. [33] proposed BABFNet, which uses cooperative learning between boundary prediction and cloud detection branches. Its semantic enhancement module supplements high-level semantic information, and its bilateral fusion module integrates multiscale and boundary guidance information, improving detection accuracy for cloud boundaries and thin clouds in complex scenes.

Since the complex geometry of clouds and limitations of conventional convolutions in the receptive field, the existing cloud detection methods are still not ideal in preserving boundary details. In addition, the existing methods generally exhibit insufficient interclass feature characterization for noncloud surface targets with similar spectral reflectance characteristics. Fig. 1 shows high albedo bare ground and static waters, which result in a high false detection rate.

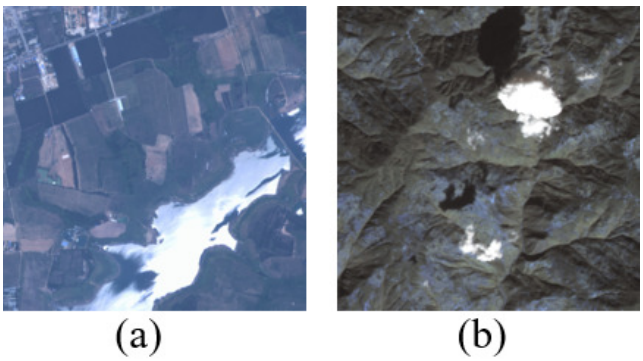


Fig. 1. During cloud detection, noncloud ground features with similar spectral reflectance characteristics can easily cause misclassification, such as (a) static waters and (b) high albedo bare ground.

To address the above problems, we propose the coarse-to-fine progressive network (CFPNet) based on a progressive feature extraction strategy. CFPNet combines the channel spatial attention residual fusion (CSARF) module and the multimask adaptive attention (MMAA) module, using the U-Net [34] architecture to achieve progressive feature extraction. The CSARF module integrates channel and SAMs. The channel attention highlights cloud-related channels and suppresses irrelevant background ones by interacting global and local channel information. The spatial attention adjusts weights of different spatial positions, enabling the network to focus on cloud areas and reducing attention to background noise. This achieves preliminary focus on cloud features in remote sensing images. The MMAA module uses a multimask self-attention module to adaptively capture long-range dependencies in cloud regions at the channel dimension. It filters channel self-attention scores through multiple masks to remove channels with weaker cloud feature associations. To achieve finer cloud and spatial features' extraction, a deformable contextual feed-forward network (DCFN) is designed, leveraging the adaptive receptive field of deformable convolutions to adjust sampling points based on cloud shapes. This strategy of progressive feature extraction in both the channel and spatial dimensions allows the network to precisely detect cloud boundaries and distinguish confusing surface features. In summary, the contributions of this article are as follows.

1) We propose CFPNet, which accurately detects clouds at boundaries and in confusing surface features by extracting features from coarse to fine.
2) We design the CSARF module. The CSARF module effectively enhances cloud features while suppressing background noise, achieving preliminary focus on cloud features.
3) We design the MMAA module. For feature maps from the CSARF module, the MMAA module further filters out channels with weak cloud feature associations and adjusts the convolutional kernel receptive field to focus on extracting cloud features. This achieves fine extraction of cloud channel and spatial dimension features.

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4) To verify the effectiveness of our proposed method, we compare it with other methods on three different datasets and conduct ablation experiments.

The rest of this article is as follows. In Section II, we discuss previous related work. In Section III, the proposed method is discussed in detail. In Section IV, the selection of datasets and experimental details are given. In Section V, detailed comparison and ablation experiments are performed, and the results are obtained and analyzed. In Section VI, the full article is summarized.

II. Related Works

A. Cloud Detection Methods With Attention Mechanisms

In the cloud detection task, features such as ice, snow, and white sand exhibit similarities in texture and color to clouds, which can lead to misjudgment during cloud detection. By simulating the selective perception characteristics of human vision, the visual attention mechanism can accurately suppress irrelevant background interference and focus on the key features of clouds. This mechanism has significant advantages in mining advanced semantic information from images. In complex scenes such as cloud detection, the visual attention mechanism dynamically allocates computational resources to enhance the model's sensitivity to subtle feature differences, thereby effectively distinguishing between clouds and confusing surface features. The attention mechanism can be divided into four main types: channel attention, spatial attention, self-attention, and hybrid attention, each of which assists the model in achieving accurate detection from different dimensions.

Channel attention focuses on the correlation between channel dimensions, concentrating the network model's attention on important channels and suppressing the interference of irrelevant channel information [35], [36]. Spatial attention dynamically adjusts the weights of different spatial locations in the feature map, aiming to focus the model on key regions in the image and suppress irrelevant background [37]. The self-attention mechanism breaks through the local receptive field limitation of the traditional convolution by calculating the correlation weights among all pixels or regions in the image, thereby achieving capture of global contextual information [38], [39], [40], [41], [42]. Researchers have also found that combining different attention mechanisms into new structures can yield good results. For example, Zhang et al. [43] proposed CFCA-Net, which uses a cascaded feature attention module (CFAM) to enhance the network's attention to important color and texture features and uses channel attention to highlight important information in the channel dimensions. Paul et al. [44] proposed a new attention structure called C-LiSA for cloud detection tasks within the PSA framework. The hierarchical cross-channel attention (HC2A) mechanism used in C-LiSA achieves good results in scenes where features such as snow, ice, and white sand match the texture and color of clouds. Du et al. [45] proposed AFMUNet, which introduces a channel spatial attention module (CSAM) to detect clouds on reflective surfaces such as snow, ice, and sand, achieving good results. Guo et al. [46] proposed CDNetV2, which uses the adaptive feature fusion model (AFFM) to improve the accuracy of cloud–snow coexistence scene recognition by adaptively fusing multilevel features of the encoder–decoder through the synergistic action of channel attention fusion model (CAFM) and spatial attention fusion model (SAFM).

B. Cloud Detection Methods With Deformable Convolution

Owing to the complexity of cloud geometry and the diversity of cloud types, it is challenging for the existing cloud detection methods to accurately delineate actual cloud regions, particularly in identifying cloud boundaries, which are susceptible to leakage and misdetection. To overcome this, researchers in recent years have developed several cloud detection methods aimed at enhancing the precision of cloud boundary detection.

Liu et al. [47] introduced DCNet, which embeds deformable convolution blocks into the shallow layers of the encoder and decoder. This allows adaptive capture of spatial contexts based on cloud shapes, thereby improving the accuracy of cloud detection results. He et al. [48] proposed DABNet, which is based on ShuffleNetV2 [49]. By using Dilated ShuffleNet (DSN) for feature extraction and replacing the original network's downsampling operations with dilation convolutions of varying rates, DABNet reduces information loss. In addition, DABNet uses a deformable contextual feature pyramid (DCFP) module to model multiscale cloud features adaptively and designs a boundary-weighted loss function to optimize boundary detection. This results in highly accurate cloud detection with clear boundaries. Hang et al. [50] proposed a dual-branch deformable convolutional network (DBDCN). It uses the Advanced Himawari-8 Imager (AHI) data and brightness temperature difference (BTD) as input. The network extracts convective cloud features and introduces a cross-branch fusion block (CFB) to fuse complementary information. This method effectively detects cloud shapes at different scales and reduces false negatives. It addresses the issues of the variable morphology of convective clouds and insufficient feature extraction. Dong et al. [51] proposed the DLACD-Net model, which adopts a nested U-shaped encoder–decoder architecture. The model incorporates depthwise separable atrous convolution residual U-block (DSACRSU) and deformable large kernel attention module to enhance multiscale cloud feature extraction and improve the accuracy of cloud detection in polar ice and snow scenes.

III. Methodology

This section provides a detailed overview of our proposed CFPNet for cloud detection. We begin by introducing our method's overall network framework, followed by an in-depth elaboration of the two key modules, CSARF and MMAA. The CSARF module consists of channel attention module (CAM) and SAM, while the MMAA module (Fig. 5) consists of MMSA and DCFN. Finally, we introduce the loss function used for training the network.

A. Overview

Fig. 2 illustrates the CFPNet architecture proposed in this article, which adopts an encoder-skip connection–decoder

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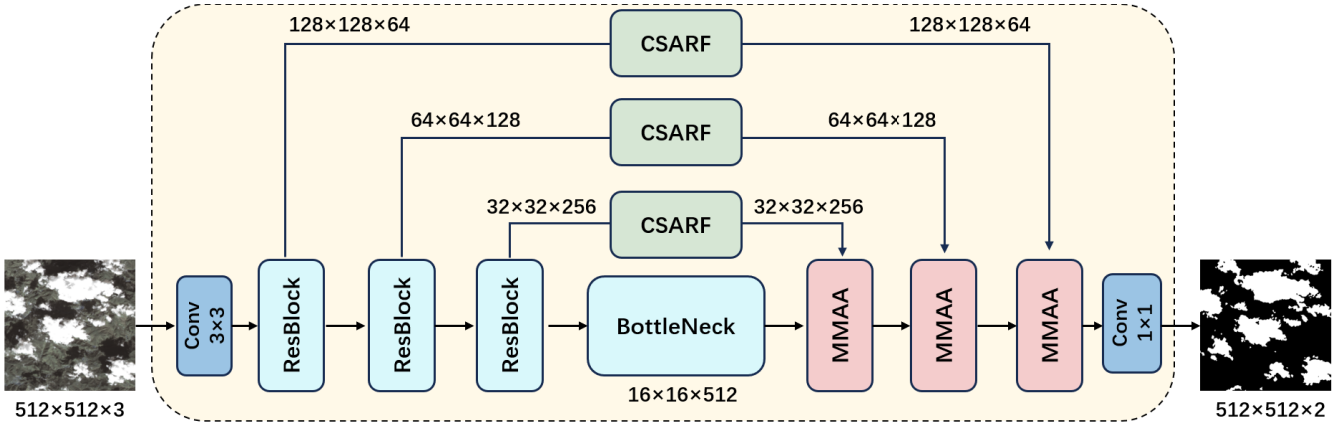


Fig. 2. Architecture of CFPNet. ResBlock is used as the encoder at each layer of the network, effectively extracting features at different levels and providing rich feature support for subsequent attention modules. The CSARF module on the skip connection is used to suppress background noise and enhance the response to cloud features, thereby achieving preliminary focus on cloud features. The MMAA module in the decoder stage fuses deep and shallow features and performs cloud feature fine extraction, and then upsamples to restore the spatial resolution of the feature map.

topology. The encoder stage adopts the residual convolutional block to construct encoder and bottleneck layer, and it provides ample feature support for the subsequent attention module. To direct the network's focus on key features, the CSARF module is designed to suppress background noise and enhance the response to cloud region features. The MMAA module is incorporated into the decoder stage. This module achieves feature refinement by integrating multilevel feature fusion of deep semantic and shallow detailed features, combined with channel refinement extraction and deformable convolution operations. Progressive upsampling is used to recover spatial resolution, and the output is a cloud detection probability map.

Specifically, the network is divided into three layers from shallow to deep, denoted as Li ($i = 1, 2, 3$). Each layer contains a set of ResBlock, CSARF, and MMAA. This section discusses the example of $L1$, I_{input} of dimension $H \times \text{W} \times 3$ is first downsampled to $(1/2)$ the size of I_{input} by a convolution operation, and then into the ResBlock to extract the feature information, and the process can be represented as

$$I_{(1/2)} = \text{ResBlock}\left(\text{Conv}_{3\times3}\left(I_{\text{input}}\right)\right) \tag{1}$$

where $\text{Conv}_{3\times3}$ denotes a convolutional kernel of size 3×3 with a step size of 2, and $I_{(1/2)}$ denotes a feature map of size $I_{\text{input}}(1/2)$ obtained by downsampling into the ResBlock, with the number of channels increasing from 3 to 64. At the skip connection, the CSARF module performs irrelevant noise suppression of the feature map $I_{(1/4)}$ from the encoder to highlight important features, a process that can be expressed as follows:

$$I_{\text{shallow}} = \text{CSARF}\left(I_{(1/2)}\right) \tag{2}$$

where I_{shallow} denotes the feature map obtained by the designed CSARF module with the same size and number of channels. The feature map I_{deep} from a deeper level is upsampled twice by linear difference and spliced with I_{shallow} for channel dimension, input to the designed MMAA module for further refinement of feature extraction, and then the output feature map is input to a 1×1 convolution kernel for full convolution

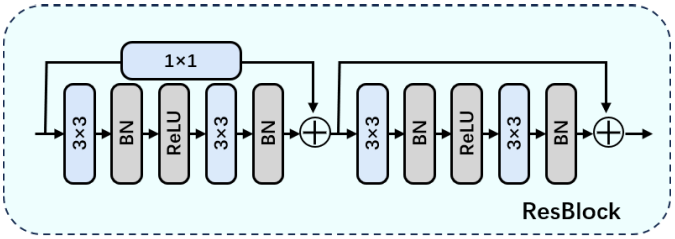


Fig. 3. Architecture of ResBlock. ResBlock extracts features and accelerates convergence through 3×3 convolution, batch normalization, and ReLU activation. The skip connection enhances feature reuse, thereby enriching feature representation.

to obtain the prediction result map. The process can be represented as

$$I_{\text{mmaa}} = \text{MMAA}\left(\text{Cat}\left[\text{Up}\left(I_{\text{deep}}\right), I_{\text{shallow}}\right]\right) \tag{3}$$

$$I_{\text{output}} = \text{Conv}_{1\times1}\left(\text{Up}\left(I_{\text{mmaa}}\right)\right) \tag{4}$$

where Up is a linear interpolation two times upsampling operation. Cat is a channel dimension concatenation operation. I_{mmaa} is a feature map of size $I_{\text{input}}(1/2)$ with 64 channels output by the MMAA module. $\text{Conv}_{1\times1}$ is a 1×1 full convolution operation that transforms the input of the feature map with 64 channels into a prediction result map with two channels.

B. Feature Encoder Module

CFPNet uses the hierarchical residual structure of pretrained ResNet-18 to construct encoders and bottleneck layer, which achieve multilevel feature representation through the migration learning mechanism. This residual network structure tackles gradient decay in deep networks with skip connection, enabling effective feature extraction while ensuring training stability. The parameter space obtained by pretraining ResNet-18 on the ImageNet [52] dataset contains versatile multilevel visual features: the shallow layer captures low-level features such as boundaries and textures, the mid-level extracts intermediate features such as geometric forms, and the deep layer

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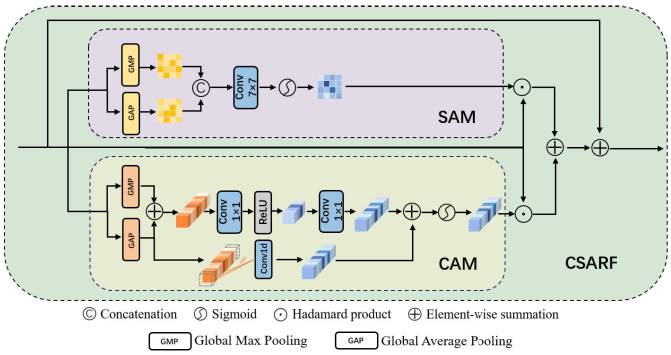


Fig. 4. Architecture of the CSARF module. The CSARF module uses CAM to highlight cloud-related channels and suppress background channels, while SAM focuses on cloud areas and reduces background noise. Skip connection is used to learn residual features.

models semantic abstract features such as classifications and the location information of the object. Compared with the random initialization strategy, the initialization distribution of the pretrained parameters is closer to the optimal solution of the remote sensing image segmentation task, which can significantly improve the model convergence efficiency.The architecture of the encoder in this article is shown in Fig. 3, which consists of two successive residual connections in series. In the first residual connection, the input feature map of the encoder is downsampled and the number of channels doubled in a step of 2 by a 1×1 convolution and a 3×3 convolution, and then the features are extracted by subsequent convolutions. This process can be represented as

$$F_{r1} = \mathrm{CB}\left(\mathrm{CBR}\left(X\right)\right) + \mathrm{Conv}_{1\times 1}\left(X\right) \tag{5}$$

$$F_{r2} = \mathrm{CB}\left(\mathrm{CBR}\left(F_{r1}\right)\right) + F_{r1} \tag{6}$$

where CBR denotes the 3×3 convolution, batch normalization, and ReLU activation function operation, CB denotes the 3×3 convolution and batch normalization operation, and F_{ri} $(i = 1, 2)$ denotes the output feature map after the ith skip connection.

C. CSARF Module

Since the encoder lacks focus on the target feature region during feature extraction, we design the CSARF to suppress extraneous background noise and highlight the target feature. The CSARF module integrates channel and spatial attention modules. The channel attention highlights cloud-related channels and suppresses irrelevant background ones by interacting global and local channel information. The spatial attention adjusts weights of different spatial positions, enabling the network to focus on cloud areas and reducing attention to background noise. By learning residual features through skip connection, the network can more accurately distinguish between cloud and background areas. Through the CSARF module, the network achieves preliminary focus on cloud features in both the channel and spatial dimensions. The architecture of the CSARF module is shown in Fig. 4.

When the input feature map X enters the CSARF module, it first goes into the SAM and the CAM. These modules generate the spatial attention matrix and the channel attention matrix, respectively. Then, through elementwise and channelwise multiplication with the input X, the module produces F_s and F_c. Finally, the input X is added element-by-element with F_s and F_c through the residual connection to get the output of the CSARF module. The result is F_{output}. The process can be expressed as follows:

$$F_s = \mathrm{SAM}\left(X\right) \odot X \tag{7}$$

$$F_c = \mathrm{CAM}\left(X\right) \odot X \tag{8}$$

$$F_{\text{output}} = F_s + F_c + X \tag{9}$$

where $\odot$ denotes the Hadamard product.

1) Channel Attention Module: As shown in Fig. 4, the context information of the input feature map X is obtained through maximum pooling and average pooling, and the spatial information is aggregated through two branches to obtain the channel attention map. One of the branches obtains the global channel interaction attention of X. The global channel information will be interacted by 1×1 convolution on the channel feature map to obtain the global channel attention map. The other branch uses dynamic 1-D convolution on the channel feature maps obtained by average pooling to achieve local channel information interaction, generating local channel attention maps. Finally, the global channel attention map and the local channel attention map are summed element by element, and then the final 1-D channel attention map is obtained by sigmoid activation function. The process can be expressed as

$$\begin{aligned} M_c = \sigma\left(\mathrm{CRC}\left(\mathrm{GAP}\left(X\right) + \mathrm{GMP}\left(X\right)\right)\right) \\ + \mathrm{Conv1D}_k\left(\mathrm{GAP}\left(X\right)\right)\end{aligned} \tag{10}$$

where GAP and GMP denote the global average pooling and global max pooling, respectively. CRC denotes the continuous convolution–ReLU–convolution operation with a convolution kernel size of 1×1. σ denotes the sigmod activation function operation. M_c denotes the 1-D channel attention map. $\mathrm{Conv1D}_k$ denotes the 1-D convolution with convolution kernel size k. The size of k is determined by the number of channels c in the input X. The expression for the relationship between k and c is

$$k = \left| \frac{\log_2\left(c\right) + 1}{2} \right|_{\text{int}} \tag{11}$$

where $|n|_{\text{int}}$ denotes an integer part that takes the absolute value of n.

2) Spatial Attention Module: As shown in Fig. 4, the module aggregates the channel information of the input feature map X through two branches, global maximum pooling and global average pooling, and then concatenates the respective spatial feature maps obtained in the channel dimension, extracts spatial local features integrating context information through 7×7 convolution, and finally obtains the 2-D spatial attention maps through the sigmoid activation function. The process can be specifically expressed as

$$M_s = \sigma\left(\mathrm{Conv}_{7\times 7}\left(\mathrm{Cat}\left[\mathrm{GAP}\left(X\right), \mathrm{GMP}\left(X\right)\right]\right)\right) \tag{12}$$

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where GAP and GMP denote global average pooling and global max pooling, respectively. σ denotes the sigmoid activation function operation. M_s denotes the 2-D spatial attention map.

D. MMAA Module

The network uses the CSARF module to achieve initial attention to target features in the channel and spatial dimensions. To further refine the extraction of cloud features in both the channel and spatial dimensions, we designed an MMAA module based on the transformer architecture at the network decoder. This module first fuses the shallow features from the CSARF module with the deep features, and then performs refined extraction of cloud features on the fused feature maps. Inspired by the sparse transformer proposed by Zhao et al. [53], the MMSA in the MMAA module uses a multimask mechanism to filter channel self-attention scores, removing channels weakly associated with cloud features, thereby reducing feature noise interactions in the standard transformer's global self-attention mechanism. In the spatial dimension, the DCFN uses the adaptive receptive field of deformable convolutions to adjust sampling points based on the shape of the cloud, thereby preserving rich cloud boundary details. The process can be specifically expressed as

$$F_{\text{fusion}} = \text{DSConv}\left(\text{Cat}\left[\text{Up}\left(F_{\text{shallow}}\right), F_{\text{deep}}\right]\right) \tag{13}$$

$$F_{l1} = F_{\text{fusion}} + \text{LN}\left(\text{MMSA}\left(F_{\text{fusion}}\right)\right) \tag{14}$$

$$F_{l2} = F_{l1} + \text{LN}\left(\text{DCFN}\left(F_{l1}\right)\right) \tag{15}$$

where LN denotes the layer normalization operation. Up denotes the twice upsampling operation. DWConv denotes the depth separable convolution operation. F_{l1} and F_{l2} denote the results after residual summation of the MMSA and DCFN modules, respectively.

Specifically, the deep features and shallow features are fused by depthwise separable convolution to obtain the fused feature map X, and then the fused feature map is convolved by depthwise separable convolution to obtain M_q, M_k, and M_v, and then M_q, M_k, and M_v are flattened from the 2-D plane of height and width into a 1-D sequence. Matrix multiplication is performed on M_q and M_K to obtain the channel self-attention matrix M_{attn}, to reduce the effect of irrelevant feature noise in M_{attn}, we use a multiscale mask Mask_i to filter the largest top k_i values of self-attention scores of each channel versus the other channels in M_{attn}, and assign the other smaller self-attention scores to 0. The self-attention matrix filtered by the masks of different scales M_{attn}^i is multiplied by the corresponding learnable weighting coefficients n_i and summed to obtain the final self-attention matrix $M_{\text{mattn}}(i = 1, 2, 3, 4)$. Finally, M_v is matrix multiplied with M_{mattn} and the result is recovered from a 1-D sequence to a 2-D plane, and then a 1×1 convolution operation is performed to obtain the output F_{MMSA} of the MMSA module. This process can be concretely represented as

$$M_q^{B\times C\times(H\cdot W)}, M_k^{B\times C\times(H\cdot W)}, M_v^{B\times C\times(H\cdot W)} = \text{Reshape}\left(\text{DWConv}\left(X\right)\right) \tag{16}$$

$$M_{\text{attn}}^{B\times C\times C} = M_q^{B\times C\times(H\cdot W)} \otimes \left[M_k^{B\times C\times(H\cdot W)}\right]^T \tag{17}$$

$$M_{\text{mattn}}^{B\times C\times C} = \sum_{i=1}^{4} \text{Softmax}\left(\text{Mask}_i\left(M_{\text{attn}}^{B\times C\times C}\right) \odot n_i\right) \tag{18}$$

$$F_{\text{MMSA}}^{B\times C\times H\times W} = \text{Reshape}\left(M_{\text{mattn}}^{B\times C\times C} \otimes M^{B\times C\times(H\cdot W)}\right) \tag{19}$$

where Reshape denotes the dimension transformation operation. $[x]^T$ denotes the transpose operation on the last two dimensions of the tensor x. n_i is uniformly initialized to 0.2. The values of k_i in Mask_i are in order $(C/2)$, $(2 \times C/3)$, $(3 \times C/4)$, $(4 \times C/5)$, and C is the number of channels of the feature map X. $\otimes$ is the matrix multiplication operation. $\odot$ is the Hadamard product operation.

After fine extraction of the channel dimension of the fused feature map X to obtain the feature map F_{l1}, to further refine the boundary of the target region of F_{l1} in the spatial dimension, we propose the DCFN module. In the DCFN module, we introduce deformable convolution. Deformable convolution overcomes the limitations of the receptive field seen in conventional convolution. It can adaptively learn offsets, allowing the convolution kernels to concentrate more closely on the target region during computation. This flexibility enables the model to capture contextual information more effectively, leading to greater accuracy in the boundary regions of detection results. As a result, it helps reduce misjudgments in cloud detection tasks. Taking the deformable convolution with a kernel size of 3×3 as an example, the receptive field size of the standard convolutional grid R can be expressed as

$$R = \begin{Bmatrix} (-1,1) & (0,-1) & (1,1) \\ (0,-1) & (0,0) & (0,1) \\ (1,-1) & (1,0) & (1,1)\,. \end{Bmatrix} \tag{20}$$

In deformable convolution, an offset $\{\Delta p_n | n = 1, \ldots, N\}$ is added to the regular convolution mesh R, where $N = |R|$, so that the deformable convolution can be expressed as

$$y\left(p_0\right) = \sum_{p_n \in R} \left(p_n\right) \cdot x\left(p_0 + p_n + \Delta p_n\right) \tag{21}$$

where p_0 denotes the position of each point on the output feature map y. p_n is an element in R that denotes the position relative to p_0. Δp_n denotes the offset learned for each position in R. It is learned by 2-D convolution, which has $2N$ channels corresponding to N 2-D offsets. Finally, bilinear interpolation is introduced to correct the offset sampled point values.

Before the deformable operation, feature maps with multiscale receptive fields are obtained by convolving the input feature maps with different-sized convolutional kernels. Then, deformable convolution is applied. This enhances the network model's ability to understand and characterize both the overall contours and local details of the target region in the image.

Specifically, the features of each channel of F_{l1} are first extracted by two different scales of depth convolution, 3×3 and 5×5, and then deformable convolution operation is performed to concatenate the feature maps obtained from the two branches in the channel dimension, and finally the feature maps are fused by 1×1 convolution to obtain the output feature map F_{DCFN} of the DCFN module. This process can be expressed as follows:

$$F_{\text{cat}} = \text{Cat}\left[\text{DCM}_{3\times3}\left(\text{DWConv}_{3\times3}\left(F_{l1}\right)\right)\right.$$

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$$\mathrm{DCM}_{3\times3}\left(\mathrm{DWConv}_{5\times5}\left(F_{l1}\right)\right)\Big] \tag{22}$$

$$F_{\mathrm{DCFN}} = \mathrm{Conv}_{1\times1}\left(F_{\mathrm{cat}}\right) \tag{23}$$

where $\mathrm{DCM}_{3\times3}$ is a deformable convolution with kernel size 3×3 and $\mathrm{DeepConv}_{k\times k}$ is a deep convolution with kernel size k. *Cat* denotes the channel dimension concatenation operation.

E. Loss Function

The cross-entropy loss function is widely used in image segmentation tasks. It compares the predicted probability of each category with the actual desired output and calculates the loss. This loss penalizes predictions based on their deviation from the true values. The goal is to minimize the loss, as a smaller loss indicates better model performance. For our network training, we selected the cross-entropy loss function, which is defined as follows:

$$L_{\mathrm{CrossEntropy}} = -\sum_{i=1}^{c} t_i \log\left(p_i\right) \tag{24}$$

where c denotes the number of predicted classes and c is equal to 2 for cloud and noncloud. t_i denotes the truth value of class i on the label. p_i denotes the probability of being class i.

IV. Dataset and Evaluation Metrics

A. Datasets

To evaluate the cloud detection performance of the proposed CFPNet, this study uses three publicly available datasets with different spatial resolutions: the GF-1 WFV cloud and cloud shadow coverage dataset, the AIR-CD cloud detection dataset, and the Sentinel-2 dataset.

The GF-1 WFV dataset, created by Li et al. [9], contains 108 class 2A images along with the corresponding cloud and cloud shadow masks. This dataset features a spatial resolution of 16 m and includes one near-infrared band and three visible bands. For our cloud detection study, we used the three visible bands. Cloud shadow pixels and background pixels are classified as noncloud pixels, noncloud pixels were set to 0, and cloud pixels were set to 255. We selected 86 original images as the training set and 22 as the test set. We performed random cropping, rotation, and flipping enhancements. All the original images were cropped into subimages of size 512×512. The training set contained 22 705 images, and the test set contained 1504 images.

The AIR-CD cloud detection dataset, created by He et al. [48], contains 34 full-scene images captured by the GF-2 satellite across various regions in China, along with reference masks. This dataset provides a higher spatial resolution of 4 m and also includes one near-infrared band and three visible bands. Compared with previous datasets, the AIR-CD dataset presents more complex and diverse scenes, making it more challenging for high-precision cloud detection. In the reference masks, cloud pixels were labeled with a value of 255, and noncloud pixels were labeled with a value of 0. For our experiments, only the three visible bands were used. In all, 25 images were randomly selected from the dataset to form the training set, and nine images were chosen for the test set. All the original images were cropped into subimages of size 512×512. The training set contained 7100 pairs, and the test set contained 1200 pairs.

The Sentinel-2 Cloud Mask Catalogue dataset was created by Aybar et al. [54] and contains 513 subscenes and reference masks. The dataset has a spatial resolution of 20 m and contains 13 bands. In the reference mask, cloud pixels were set to 255, and noncloud pixels were set to 0. In the experiments, remote sensing images with 13 bands were used as input to test the cloud detection performance of each method on multiband remote sensing images. A training set of 450 subscenes were randomly selected from the Level-1C Sentinel-2 dataset, and the remaining 63 subscenes were used as the test set. All the original images were cropped into subimages of size 256×256. The training set contained 7100 pairs, and the test set contained 1008 pairs.

B. Evaluation Metrics

To evaluate the performance of different methods in remote sensing image cloud detection, we use five widely used quantitative metrics, including overall accuracy (OA), precision, recall, F1-score, and mean intersection over union (MIoU). It is worth noting that MIoU, OA, and F1 measure the combined performance of the models, and the larger the value, the better the overall performance, and their formulae are as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{25}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{26}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{27}$$

$$\mathrm{F1\text{-}Score} = \frac{2\times\mathrm{Precision}\times\mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}} \tag{28}$$

$$\mathrm{MIoU} = \frac{TP}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}}. \tag{29}$$

TP, TN, FP, and FN denote a correct prediction of a cloud, a correct prediction of a noncloud, a false positive, and a false negative, respectively. OA, F1-score, and MIoU are comprehensive evaluation metrics that provide a more accurate assessment of overall network performance.

V. Experimental Results and Discussion

A. Experimental Setup and Methods of Comparison

CFPNet is implemented on the PyTorch DL framework. The network is trained on a single NVIDIA RTX 4090 GPU. The optimizer used in the experiment is Adam, with a learning rate of 0.00001. The batch size is set to 8. The maximum number of training iterations is set to 100. To evaluate the performance of the proposed CFPNet in detecting cloud boundaries and distinguishing spectrally similar surfaces, we choose the semantic segmentation methods DeepLabV3+ [19], DANet [41], and UNetFormer [55] and cloud detection methods CDNetV2 [46], DCNet [47], BoundaryNet [32], BABFNet [33], and SwinCloud [56] for comparison. The experimental environments and experimental settings of all the above compared methods are consistent with the methods in this article.

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TABLE I

COMPARISON OF QUANTITATIVE RESULTS OF THE METHODS IN THIS ARTICLE TO DIFFERENT METHODS ON GF1-WFV ASSESSMENT TEST DATA. FIRST, SECOND, AND THIRD METHODS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Method	Year	OA(%)	Precision(%)	Recall(%)	F1-Score(%)	MIoU(%)
DeepLabV3+ [19]	2018	**95.77**	**97.46**	92.21	**94.76**	**90.04**
DANet [41]	2019	95.48	94.98	**94.07**	94.52	89.62
CDNetV2 [46]	2021	**95.64**	95.90	93.47	**94.67**	**89.88**
DCNet [47]	2022	94.79	96.26	90.97	93.54	87.86
UNetFormer [55]	2022	95.03	94.30	93.68	93.99	88.67
BoundaryNet [32]	2022	95.51	94.49	**94.68**	94.59	89.73
BABFNet [33]	2023	95.31	**96.34**	92.20	94.22	89.08
SwinCloud [56]	2024	95.03	94.66	93.28	93.96	88.62
CFPNet	–	**96.53**	**96.50**	**95.07**	**95.78**	**91.90**

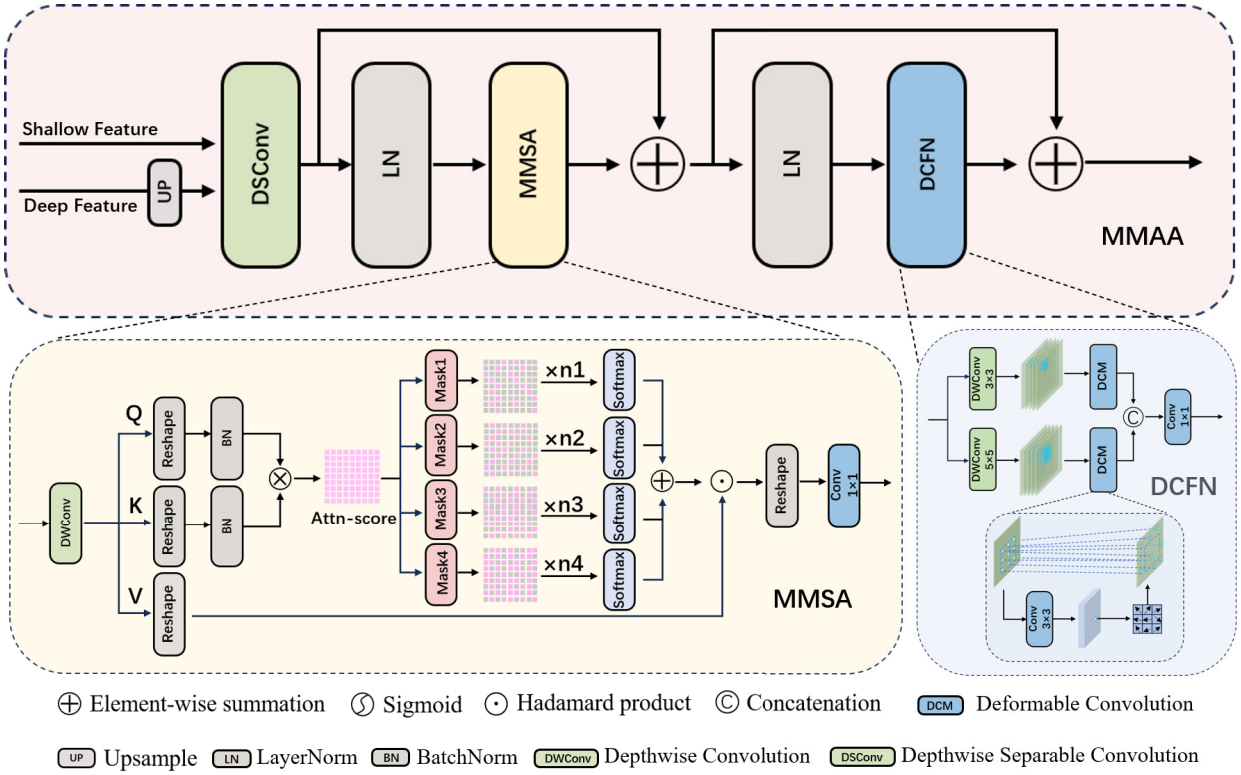


Fig. 5. Architecture of the MMAA module. Depth separable convolutions fuse deep features and shallow features. In the channel dimension, the MMSA uses a multimask mechanism to filter out channels weakly associated with cloud features, reducing feature noise interactions. In the spatial dimension, the DCFN leverages deformable convolutions' adaptive receptive field to adjust sampling points based on cloud shapes, preserving rich cloud boundary details.

B. Cloud Detection Results on the GF1-WFV Dataset

Experiments on the GF1-WFV dataset show that the proposed CFPNet method significantly outperforms the existing benchmark methods. Table I summarizes the quantitative results of the methods on the GF1-WFV dataset, and although the performance on the precision metrics is not the best, the proposed CFPNet achieves the best performance on the comprehensive metrics OA, F1-score, and MIoU.

The visualization results in Fig. 6 further demonstrate that CFPNet exhibits excellent geometric detail retention capabilities in cloud scenes with diverse shapes. Specifically, in thick cloud scenes, as shown in Fig. 6(1) and (2), the SAM and deformable convolution effectively improve the accuracy of cloud boundary detection. DeepLabV3+ [19] performs context modeling through multiscale dilated convolutions, but its fixed receptive field and loss of local information prevent it from fully restoring boundary details. Although DANet [41] enhances feature representation through a dual attention mechanism, the lack of cross-layer connections complicates the effective utilization of shallow geometric features, resulting in blurred boundaries in detection results. DCNet [47] enhances boundary detection precision by cross-layer feature fusion and uses deformable convolution layers to better adapt to cloud shapes. BoundaryNet [32] and BABFNet [33] optimize boundary detection through dedicated boundary compensation branches, but multiscale dilated convolution operations in boundary regions can lead to feature information loss, causing false detections. In the scenes where thick and thin

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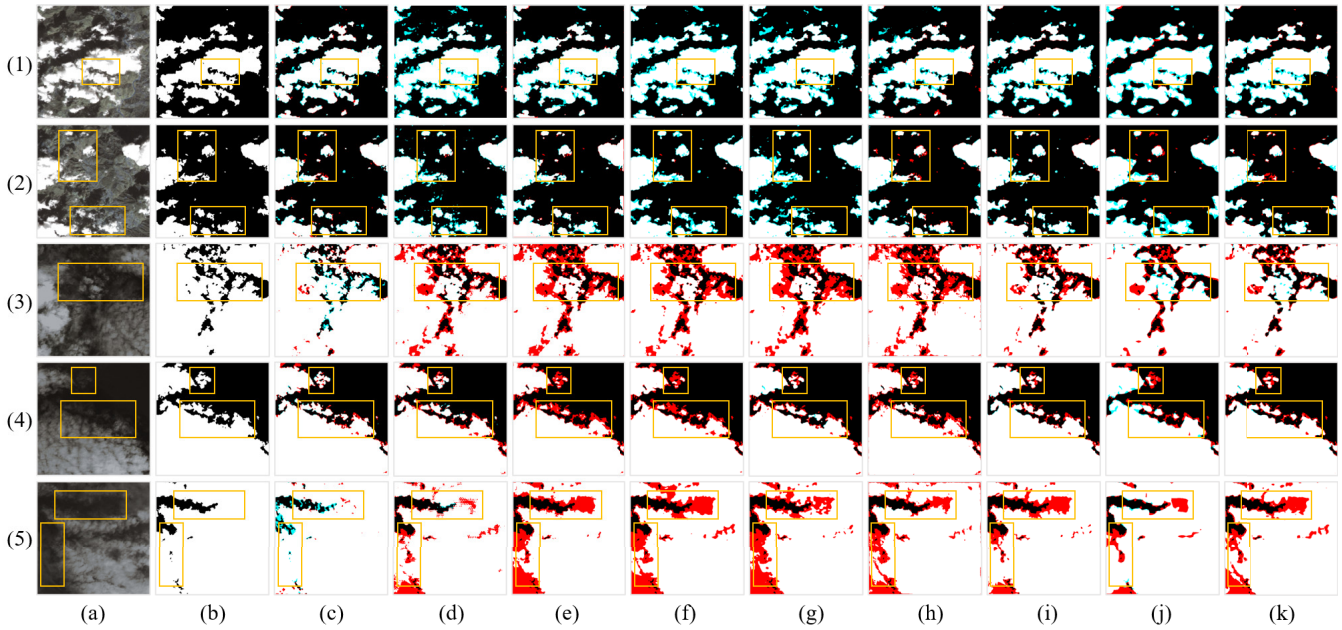


Fig. 6. Visual comparison of different methods in five example scenes of the GF1-WFV dataset. The blue part is false detection, and the red part is missed detection. (a) False-color RSI (image). (b) Ground truth. (c) CFPNet. (d) SwinCloud. (e) BABFNet. (f) BoundaryNet. (g) UNetFormer. (h) DCNet. (i) CDNetV2. (j) DANet. (k) DeepLabV3+.

TABLE II

COMPARISON OF QUANTITATIVE RESULTS OF THE METHODS IN THIS ARTICLE TO DIFFERENT METHODS ON AIR-CD ASSESSMENT TEST DATA. FIRST, SECOND, AND THIRD METHODS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Method	Year	OA(%)	Precision(%)	Recall(%)	F1-Score(%)	MIoU(%)
DeepLabV3+ [19]	2018	**96.63**	**88.83**	83.22	**85.93**	**75.34**
DANet [41]	2019	96.49	86.20	**85.72**	**86.91**	75.00
CDNetV2 [46]	2021	**96.56**	**89.14**	82.18	85.52	74.70
DCNet [47]	2022	96.30	85.56	84.26	84.91	73.77
UNetFormer [55]	2022	96.09	85.04	82.95	83.98	72.38
BoundaryNet [32]	2022	96.26	86.50	82.67	84.54	73.22
BABFNet [33]	2023	96.55	87.06	**84.69**	85.85	**75.20**
SwinCloud [56]	2024	96.11	87.13	80.41	83.64	71.87
CFPNet	–	**97.02**	**87.76**	**88.24**	**88.00**	**78.57**

clouds coexist in Fig. 6(3)–(5), CFPNet enables global–local channel information interaction through the CAM in skip connections, and models thin cloud features through multimask channel self-attention in decoders, significantly improving thin cloud detection accuracy. Compared with CDNetV2 [46] and DANet [41], which use simple channel attention mechanisms, our method achieves more precise feature selection through coarse-to-fine attention mechanisms, thereby significantly reducing the false-negative rate. Although SwinCloud [56] and UNetFormer [55] use window self-attention to extract features, their receptive fields are limited to the window size, making it difficult to fully capture the weak features of thin clouds and leading to false negatives. The experimental results confirm that the proposed method performs effectively in preserving boundary details in both thick and thin cloud scenes.

C. Cloud Detection Results on the AIR-CD Dataset

To evaluate the performance of the proposed CFPNet in various scenes, we choose the AIR-CD cloud detection dataset for training and testing, whose higher spatial resolution and diverse scenes are more challenging for each method. Table II summarizes the quantitative results of each method on the AIR-CD cloud detection dataset. Although the performance on the precision metrics is not the best, the proposed CFPNet achieves the best performance on the comprehensive metrics OA, F1-score, and MIoU.

In the thick cloud scenes shown in Fig. 7(1) and (2), the CFPNet achieves more accurate results using a spatial dimension progressive cloud feature extraction strategy. BABFNet [33] and BoundaryNet [32] perform well in capturing boundary details within cloud regions, thanks to their cloud boundary prediction modules, which provide rich boundary detail information. Other methods accurately predict the overall cloud regions but lose boundary detail information to varying degrees. In the thin cloud scenes shown in Fig. 7(3), the CFPNet successfully detects most thin cloud regions. Other comparison methods exhibit varying degrees of false negatives. Notably, while DANet [41] can detect most thin cloud regions, it lacks boundary details, as large-scale

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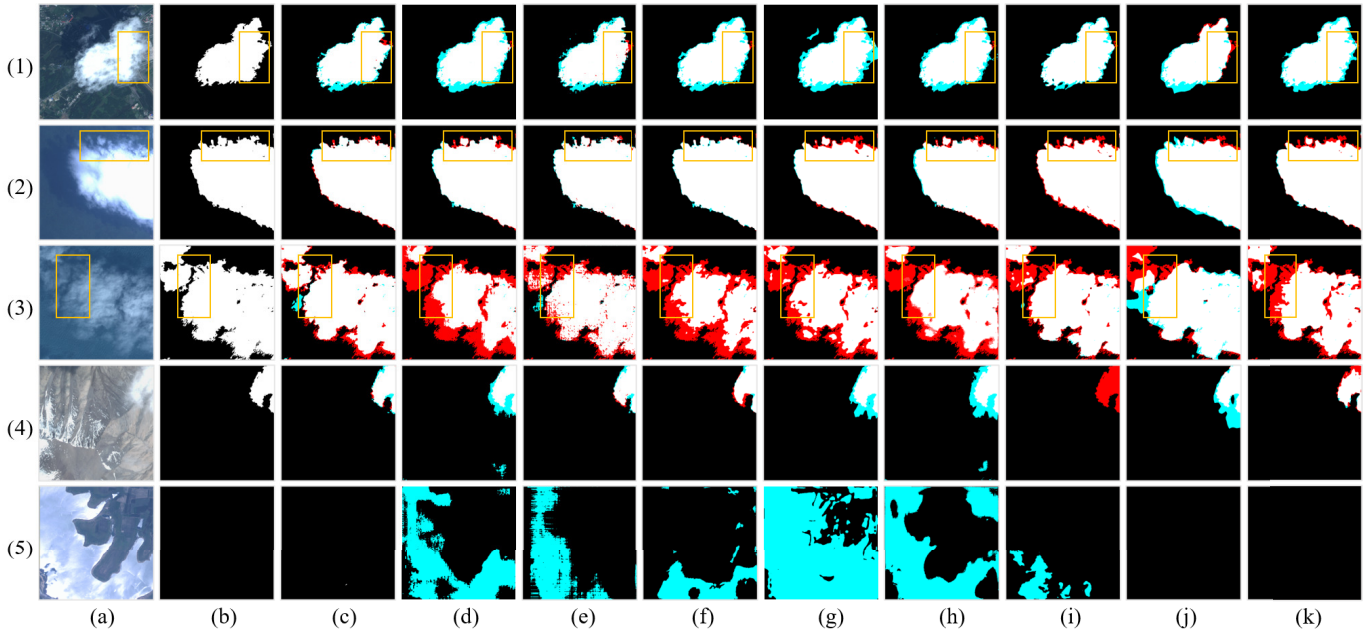


Fig. 7. Visual comparison of the different methods in five example scenes of the AIR-CD dataset. The blue part is false detection, and the red part is missed detection. (a) False-color RSI (image). (b) Ground truth. (c) CFPNet. (d) SwinCloud. (e) BABFNet. (f) BoundaryNet. (g) UNetFormer. (h) DCNet. (i) CDNetV2. (j) DANet. (k) DeepLabV3+.

TABLE III

COMPARISON OF QUANTITATIVE RESULTS OF THE METHODS IN THIS ARTICLE TO DIFFERENT METHODS ON SENTINEL-2 ASSESSMENT TEST DATA. FIRST, SECOND, AND THIRD METHODS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE, RESPECTIVELY

Method	Year	OA(%)	Precision(%)	Recall(%)	F1-Score(%)	MIoU(%)
DeepLabV3+ [19]	2018	94.97	96.73	94.48	95.59	91.55
DANet [41]	2019	95.09	**97.04**	94.36	95.68	91.72
CDNetV2 [46]	2021	95.14	95.22	**96.40**	**95.81**	**91.96**
DCNet [47]	2022	94.15	95.36	94.45	94.90	90.30
UNetFormer [55]	2022	94.30	96.24	93.78	94.99	90.46
BoundaryNet [32]	2022	**95.22**	96.36	**95.31**	**95.83**	**92.00**
BABFNet [33]	2023	**95.16**	**96.74**	94.80	95.76	91.87
SwinCloud [56]	2024	94.48	95.45	94.95	95.20	90.84
CFPNet	–	**96.19**	**96.90**	**96.47**	**96.68**	**93.58**

upsampling operations result in the loss of detail information. SwinCloud [56] and UNetFormer [55] exhibited significant missed detection. The use of window self-attention may result in the loss of global contextual information, thereby affecting the capture of cloud features. In the snow and bright lake surface scenes shown in Fig. 7(4) and (5), the proposed CFPNet effectively distinguishes between spectrally similar surface features and clouds, with more accurate prediction of cloud boundary details. In contrast, other comparison methods exhibit varying degrees of severe missed detection and false detection. This is attributed to CFPNet's progressive feature extraction strategy in the channel dimension, which enhances cloud-related feature channels while reducing the influence of noncloud background feature channels. In contrast, other methods have limitations in distinguishing cloud features from similar features. Analysis of the experimental results indicates that the proposed CFPNet can effectively distinguish clouds from confusing surface features and performs exceptionally well in thin clouds and boundary details.

D. Cloud Detection Results on the Sentinel-2 Dataset

To evaluate the effectiveness of the proposed CFPNet for cloud detection in multiband remote sensing images, we used the Sentinel-2 dataset for training and testing. The complex and diverse scenes in this dataset challenge the ability of each method to effectively use information from each band for cloud detection. Table III summarizes the quantitative results of each method on the Sentinel-2 cloud detection dataset. The proposed CFPNet adopts a progressive channel extraction strategy, which can fully extract and highlight the important channels of cloud in multiband remote sensing images, thus achieving more competitive performance.

In the thick cloud scene shown in Fig. 8(1), the CFPNet effectively detects the cloud and retains relatively accurate boundary details. BABFNet [33] and BoundaryNet [32] retain boundary details but exhibit partial false negatives. Other methods show varying degrees of false negatives, with boundary details appearing overly smooth and blurred. In the large-scale thin cloud and small-scale thin cloud scenes shown

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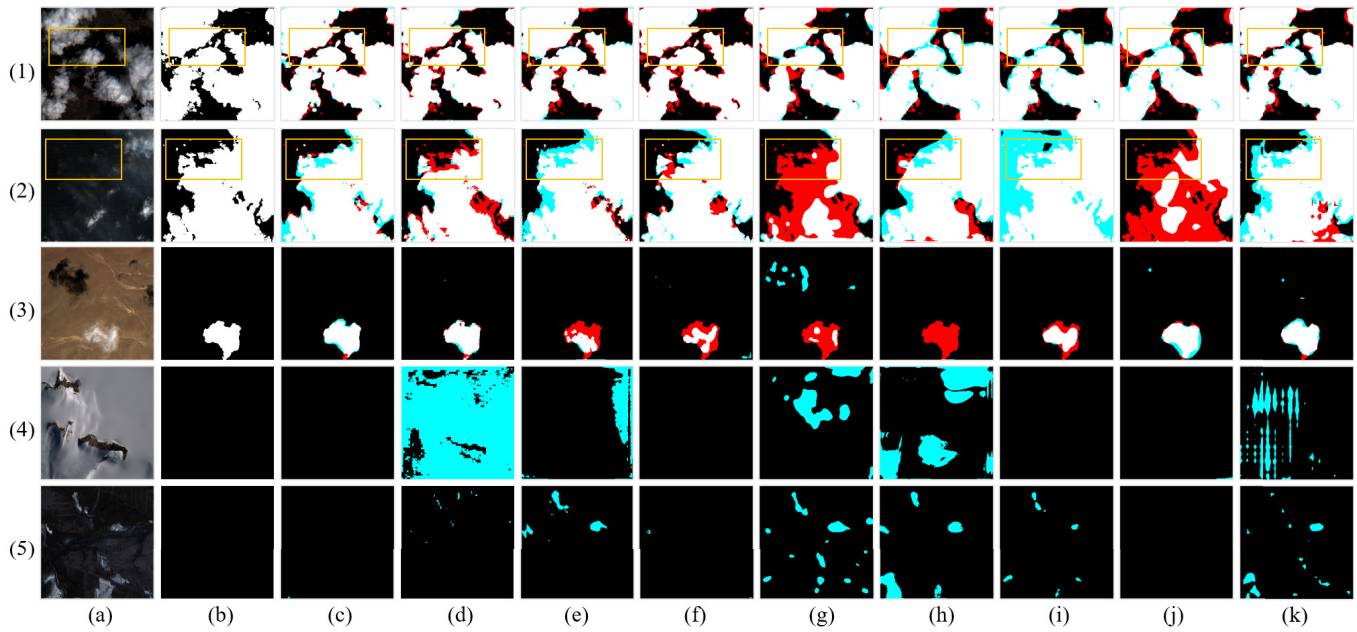


Fig. 8. Visual comparison of different methods in five example scenes of the Sentinel-2 dataset. The blue part is false detection, and the red part is missed detection. (a) False-color RSI (image). (b) Ground truth. (c) CFPNet. (d) SwinCloud. (e) BABFNet. (f) BoundaryNet. (g) UNetFormer. (h) DCNet. (i) CDNetV2. (j) DANet. (k) DeepLabV3+.

in Fig. 8(2) and (3), CFPNet effectively detects thin cloud with high accuracy. This demonstrates the effectiveness of the collaborative extraction of spatial features through the combination of spatial attention and deformable convolutions. Other methods exhibit varying degrees of false positives and false negatives. Notably, SwinCloud [56] also achieves good results in small-scale thin cloud scenes, demonstrating the effectiveness of using window self-attention as an encoder for local thin cloud feature extraction. Although DANet [41] and DeepLabV3+ [19] can also effectively detect small-scale thin clouds, they suffer from boundary blurring due to large-scale upsampling operations. In the snowy and high-brightness ground surface building scenes shown in Fig. 8(4) and (5), the CFPNet demonstrates better adaptability and robustness, attributed to its progressive channel dimension feature extraction method, which fully learns the channel representations of clouds and nonclouds. Based on the performance across these scenes, the proposed CFPNet demonstrates greater competitiveness in cloud detection on multiband remote sensing images, particularly in boundary details and distinguishing confusing surface features.

E. Computational Complexity Analysis

In Table IV, we used three evaluation metrics to evaluate the computational complexity of these networks: floating point operations (FLOPs), parameters, and inference time. FLOPs and inference time are obtained using input data of size 512 $\times$ 512. It can be seen that CFPNet has a higher number of parameters than BABFNet [33], UNetFormer [55], and DCNet [47]. BABFNet [33] effectively controls the number of parameters through its shallow boundary prediction branch and parameter sharing design. UNetFormer [55]'s lightweight design is primarily due to its use of a ResNet-18 encoder and an optimized transformer decoder, which allow the model to maintain high efficiency while significantly reducing the number of parameters. DCNet [47], based on U-Net [34], adds deformable convolution layers in the shallow layers of the encoder and decoder, resulting in a smaller parameter count and computational complexity. Notably, CFPNet has fewer parameters and faster computational efficiency than the transformer-based SwinCloud [56]. In summary, CFPNet performs well in computational and parameter complexity but could be improved in inference speed, showing balanced performance compared with other methods.

TABLE IV
COMPUTATIONAL COMPLEXITY ANALYSIS OF DIFFERENT METHODS

Method	GFLOPs (1G = 1×10^9)	Params (1M = 1×10^6)	Time (image/ms)
DeepLabV3+ [19]	64.64	15.31 M	9.1
DANet [41]	57.20	47.43 M	17.8
CDNetV2 [46]	67.06	67.08 M	40.8
DCNet [47]	23.38	2.85 M	9.7
UNetFormer [55]	11.74	11.68 M	16.1
BoundaryNet [32]	510.40	88.89 M	37.3
BABFNet [33]	62.69	3.49 M	25.0
SwinCloud [56]	84.60	76.32 M	52.0
CFPNet	47.81	12.24 M	20.4

F. Ablation Experiments

To evaluate the effectiveness of each module of CFPNet, ablation experiments are conducted on the GF1-WFV dataset using OA, precision, recall, F1-score, and MIoU as evaluation metrics. Table V presents the performance comparison of

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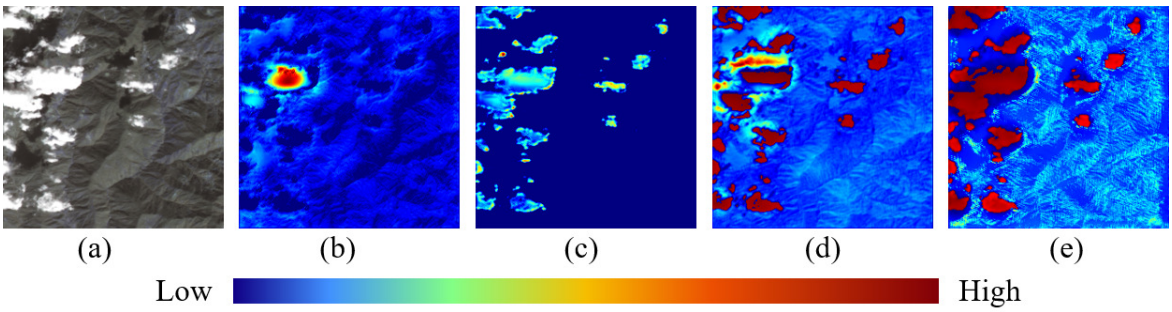


Fig. 9. Heat maps of ablation for different modules. (a) False-color image. (b) Baseline. (c) Baseline + CSARF. (d) Baseline + MMAA. (e) Baseline + CSARF + MMAA.

TABLE V
COMPARISON RESULTS OF MASK COMBINATIONS FOR DIFFERENT MODULES. BEST RESULTS ARE IN BOLD

Method	OA(%)	Precision(%)	Recall(%)	F1-Score(%)	MIoU(%)
Baseline	96.05	**97.31**	93.05	95.13	90.72
Baseline+CSARF	96.33	96.56	94.52	95.53	91.44
Baseline+MMAA	96.36	96.61	94.54	95.56	91.51
Baseline+CSARF+MMAA	**96.53**	96.50	**95.07**	**95.78**	**91.90**

TABLE VI
COMPARISON RESULTS OF MASK COMBINATIONS FOR DIFFERENT SCALE RANGES. BEST RESULTS ARE IN BOLD

Range	OA(%)	Precision(%)	Recall(%)	F1-Score(%)	MIoU(%)
[1/20,1/4]	96.12	95.80	94.80	95.29	91.01
[2/7,9/20]	96.35	96.36	94.77	95.56	91.50
[1/2,4/5]	**96.53**	**96.50**	95.07	**95.78**	**91.90**
[5/6,9/10]	96.42	95.95	**95.39**	95.67	91.69
[23/25,1]	96.17	95.86	94.86	95.36	91.14

different modules, Table VI shows the effect of different mask ranges in MMSA on network performance, and Fig. 9 displays the heat maps of different modules for detecting cloud regions.

1) Ablation Experiments for Each Module of the Model: The results in Table V indicate that the complete framework of Baseline + CSARF + MMAA achieves optimal performance in comprehensive metrics such as OA, F1-score, and MIoU. The precision of the complete CFPNet is lower than that of other combinations because the other combinations' networks are too conservative and can only correctly detect clouds within a small range, thus avoiding excessive false positives. This results in a significant increase in precision and a significant decrease in recall. This situation is most evident in the baseline combination. Through the integrated analysis of precision and recall, this framework effectively combines the feature enhancement capability of the CSARF module and the detail optimization characteristics of the MMAA module, achieving synergistic optimization in cloud area positioning accuracy and coverage completeness.

The visualization in Fig. 9 elucidates the role of each module. Fig. 9(a) shows that the baseline model can only detect local cloud areas due to the lack of an attention mechanism, resulting in limited context modeling capability. Fig. 9(c) demonstrates that introducing the CSARF module significantly improves the coverage of cloud regions through the channel SAM. Fig. 9(d) reveals that when only the MMAA module is introduced, its fine-grained channel self-attention mechanism extends the feature attention range, while the deformable convolutional feedforward network (DCFN) effectively captures boundary features. Fig. 9(e) illustrates that the complete network, through a coarse-to-fine strategy, achieves progressive feature extraction. The CSARF module provides prior feature focus on the channel and spatial dimensions, and the MMAA module further refines feature extraction, ultimately balancing accuracy and detail.As shown in Fig. 10, each module enhances the network's attention to cloud regions. In Fig. 10(1) and (2), thick cloud scenes are shown. Without the attention mechanism, the baseline network exhibits a degree of missed detection. By integrating the CSARF module's initial cloud focus and the MMAA module's refined feature extraction, the network's cloud detection capabilities are significantly improved. Furthermore, in the coexistence scenes of thick and thin clouds illustrated in Fig. 10(3)–(5), each module continues to effectively reduce missed detections. The complete network, leveraging the synergistic effects of two modules, achieves the optimal performance in cloud detection. These experimental results confirm the complementary optimization mechanism of the two modules. The CSARF module enhances focus on the cloud regions by suppressing background noise, while the MMAA module

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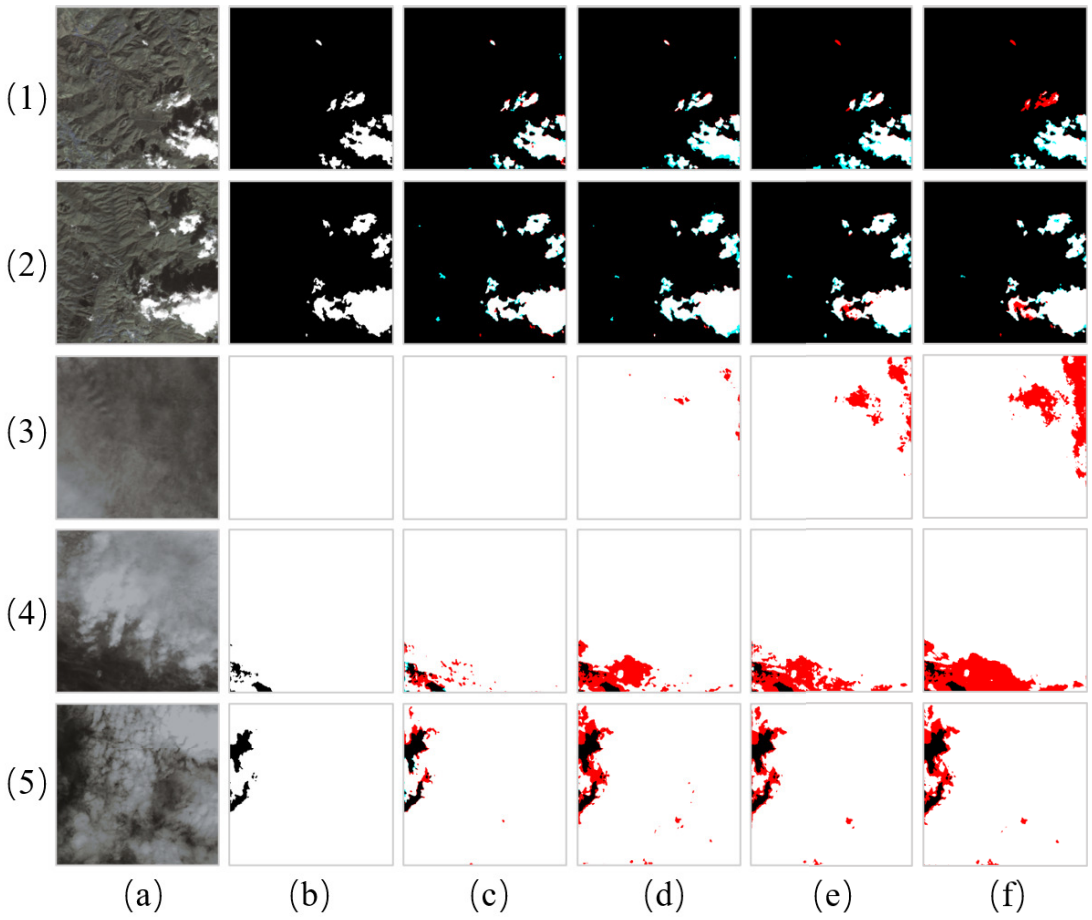


Fig. 10. Visual results of ablation study for different modules. The blue part is false detection, and the red part is missed detection. (a) False-color RSI (image). (b) Ground truth. (c) Baseline + CSARF + MMAA (ours). (d) Baseline + MMAA. (e) Baseline + CSARF. (f) Baseline.

improves boundary characterization through long-range dependencies modeling and local adaptive modeling. The synergy between them enables the model to achieve boundary detail reconstruction while maintaining high detection accuracy.

2) Ablation Experiments on the Range of Scale Combinations With Multimask: As shown in Table VI, it can be seen that the combination range [1/2, 4/5] we selected performed best in terms of the comprehensive evaluation indicators OA, F1-score, and MIoU. This combination range enables the model to focus on key channels while effectively reducing background noise, thereby improving cloud detection accuracy. As the combination range gradually deviates from the optimal combination range, the performance of the network gradually decreases. This is because the combination range decreases, while reducing more background noise, also reducing attention to important features, resulting in incomplete cloud detection regions. Conversely, as the combination range increases, while it can expand the scope of capturing important features, it also introduces excessive background noise features into the model, resulting in cloud feature representation containing too much irrelevant information, increasing the risk of false positives and false negatives, and thereby reducing the overall detection accuracy.

VI. Conclusion

To enhance the cloud detection of cloud boundaries and confusing regions, this article proposes the CFPNet which adopts a coarse-to-fine feature extraction strategy. CFPNet extracts rich features using the pretrained ResNet-18 model as the encoder. A CSARF module is designed at the skip connection to initially focus on the cloud regions and suppress background noise. The MMAA module, incorporated into the decoder, fuses deep contextual semantic features with shallow detailed features. By establishing adaptive long-range dependencies and local adaptive modeling, it enables fine extraction of cloud regions in both channel and spatial dimensions. To evaluate the effectiveness of the method proposed in this article, we selected several representative semantic segmentation and cloud detection methods to conduct experiments on the GF1-WFV, AIR-CD, and Sentinel-2 datasets. The results of these experiments demonstrate that our method gets better results. Based on the experimental results from the Sentinel-2 dataset, we can attempt to use CFPNet to obtain cloud detection results from optical images as prior information in the cloud removal task based on the SEN12MS-CR dataset [57]. This information can then be used to guide the cloud removal network in reconstructing information for cloudy regions while maximizing the retention of spectral information in cloud-free

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regions, thereby improving the quality of the reconstructed images. However, we also note that the proposed network is still not sufficiently lightweight, making it challenging to deploy in resource-constrained environments. In future work, we aim to explore lightweight, high-precision networks and investigate more effective cloud removal possibilities when combined with the cloud removal network.

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